

Homework 1

Solution

1. Compute each of the following limits or prove that it doesn't exist in the extended sense

(a) $\lim_{x \rightarrow 0} \frac{x^2 - x \sin x}{(e^x - 1) \cdot \ln(1+x)}.$

Solution:

First note that

$$\lim_{x \rightarrow 0} \frac{x}{e^x - 1} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{1}{e^x} \stackrel{\text{AOL}}{=} 1 \quad (\star)$$

and

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{\ln(1+x)} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{1 - \cos x}{\left[\frac{1}{1+x}\right]} \stackrel{\text{AOL}}{=} 0 \quad (\star\star)$$

It follows that

$$\lim_{x \rightarrow 0} \frac{x^2 - x \sin x}{(e^x - 1) \cdot \ln(1+x)} = \lim_{x \rightarrow 0} \frac{x(x - \sin x)}{(e^x - 1) \cdot \ln(1+x)} = \lim_{x \rightarrow 0} \left[\frac{x}{e^x - 1} \cdot \frac{x - \sin x}{\ln(1+x)} \right] \stackrel{(\star), (\star\star)}{=} 0$$

(b) $\lim_{x \rightarrow 0^+} [x^2 \cdot \ln(x)].$

Solution:

We have

$$\lim_{x \rightarrow 0^+} [x^2 \cdot \ln(x)] = \lim_{x \rightarrow 0^+} \frac{\ln(x)}{\left[\frac{1}{x^2}\right]} \stackrel{\frac{-\infty}{\infty}}{=} \lim_{x \rightarrow 0^+} \frac{\left[\frac{1}{x}\right]}{\left[-\frac{2}{x^3}\right]} = \lim_{x \rightarrow 0^+} \left[-\frac{x}{2}\right] \stackrel{\text{AOL}}{=} 0$$

(c) $\lim_{x \rightarrow 0} \frac{\tan(x) - x}{\ln(1+x) - x}.$

Solution:

we have

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\tan(x) - x}{\ln(1+x) - x} &\stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{\frac{1}{\cos^2(x)} - 1}{\frac{1}{1+x} - 1} = \lim_{x \rightarrow 0} \frac{\left[\frac{1 - \cos^2(x)}{\cos^2(x)}\right]}{\left[\frac{1 - (1+x)}{1+x}\right]} \stackrel{\text{formula}}{=} \lim_{x \rightarrow 0} \frac{\sin^2(x) \cdot (1+x)}{\cos^2(x) \cdot (-x)} \\ &= \lim_{x \rightarrow 0} \left[-\frac{\sin(x)}{x} \cdot \sin(x) \cdot \frac{1}{\cos^2(x)} \cdot (1+x) \right] \stackrel{\text{AOL}}{=} 0 \end{aligned}$$

(d) $\lim_{x \rightarrow \infty} (e^{2x} - x + 1)^{\frac{1}{x}}.$

Solution:

$$\lim_{x \rightarrow \infty} (e^{2x} - x + 1)^{\frac{1}{x}} = \lim_{x \rightarrow \infty} e^{\ln \left[(e^{2x} - x + 1)^{\frac{1}{x}} \right]} = \lim_{x \rightarrow \infty} e^{\frac{\ln(e^{2x} - x + 1)}{x}} = (\star)$$

Let us calculate the limit

$$\lim_{x \rightarrow \infty} \frac{\ln(e^{2x} - x + 1)}{x} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{\left[\frac{e^{2x} \cdot 2 - 1}{e^{2x} - x + 1} \right]}{1} = \lim_{x \rightarrow \infty} \frac{e^{2x} \cdot 2 - 1}{e^{2x} - x + 1} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{4 \cdot e^{2x}}{2 \cdot e^{2x} - 1} \stackrel{\frac{\infty}{\infty}}{=} \lim_{x \rightarrow \infty} \frac{8 \cdot e^{2x}}{4 \cdot e^{2x}} = \lim_{x \rightarrow \infty} 2 = 2$$

We conclude that

$$(\star) = \lim_{x \rightarrow \infty} e^{\frac{\ln(e^{2x} - x + 1)}{x}} \stackrel{\text{AOL}}{=} e^2$$

(e) $\lim_{x \rightarrow 0} \frac{(x+e)^x - e^x}{x^2}.$

Solution:

By AOD for every $x > -e$, $(x+e)^x$ is differentiable and

$$((x+e)^x)' = \left(e^{x \ln(x+e)} \right)' = (x+e)^x \cdot \left(\ln(x+e) + \frac{x}{x+e} \right)$$

Now

$$\lim_{x \rightarrow 0} \frac{(x+e)^x - e^x}{x^2} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0} \frac{(x+e)^x \cdot \left(\ln(x+e) + \frac{x}{x+e} \right) - e^x}{2x} \stackrel{\frac{0}{0}}{=}$$

$$\lim_{x \rightarrow 0} \frac{(x+e)^x \cdot \left(\ln(x+e) + \frac{x}{x+e} \right)^2 + (x+e)^x \cdot \left(\frac{1}{x+e} + \frac{e}{(x+e)^2} \right) - e^x}{2} \stackrel{\text{AOL}}{=}$$

$$\frac{1 \cdot (1+0)^2 + 1 \cdot \left(\frac{1}{e} + \frac{1}{e} \right) - 1}{2} = \frac{\left(\frac{2}{e} \right)}{2} = \frac{1}{e}$$

2. Let $x_0 \in \mathbb{R}$ and let f be a function that is defined on a neighborhood of x_0 . Suppose that f is differentiable at every x in the neighborhood for which $x \neq x_0$. Let $L \in \mathbb{R}$ and suppose also that $\lim_{x \rightarrow x_0} f'(x) = L$.

- (a) Give an example of a function satisfying the conditions above, and such that f is not differentiable at x_0 .

Solution:

Consider the function defined by $f(x) = \begin{cases} 0 & x \neq 0 \\ 1 & x = 0 \end{cases}$ for every $x \in \mathbb{R}$. We choose $x_0 = 0$.

Note that for every $x \neq 0$, f is differentiable by AOD at x and that for every $x \neq 0$ we have $f'(x) = 0$.

It follows that

$$\lim_{x \rightarrow 0} f'(x) = \lim_{x \rightarrow 0} 0 = 0$$

Note also that $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} 0 = 0 \neq 1 = f(0)$, which implies that f is not continuous at $x_0 = 0$ and therefore f is not differentiable at $x_0 = 0$.

- (b) Suppose, in addition, that f is continuous at x_0 . Prove that f is continuously differentiable at x_0 .

Solution:

As f is continuous at x_0 , then $\lim_{x \rightarrow x_0} f(x) = f(x_0)$. It follows that

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] \stackrel{\text{AOL}}{=} 0$$

Thus, by L'Hôpital's rule

$$f'(x_0) = \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow x_0} \frac{f'(x)}{1} = \lim_{x \rightarrow x_0} f'(x) = L$$

It follows that f is continuously differentiable at x_0 and that $f'(x_0) = L = \lim_{x \rightarrow x_0} f'(x)$.

- (c) Conclude that there exists no function $f : (-1, 1) \rightarrow \mathbb{R}$ for which $f'(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 1 & x = 0 \end{cases}$ for every $x \in (-1, 1)$.

Solution:

ATC that such a function exists. As f is differentiable at $x_0 = 0$ then f is continuous at $x_0 = 0$.

As $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} [x \sin \frac{1}{x}] \stackrel{B \times V}{=} 0$, then by item (b), f is continuously differentiable at $x_0 = 0$ and therefore

$$1 = f'(0) = \lim_{x \rightarrow 0} f'(x) = 0$$

Contradiction.

3. Let $x_0 \in \mathbb{R}$ and let $f : \mathbb{R} \rightarrow \mathbb{R}$.

(a) Suppose that f is twice-differentiable at x_0 . Prove that

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2} = f''(x_0)$$

Solution:

As f is twice differentiable, then f is continuous and therefore $\lim_{h \rightarrow 0} f(x_0 + h) = f(x_0)$ and $\lim_{h \rightarrow 0} f(x_0 - h) = f(x_0)$.

It follows that

$$\lim_{h \rightarrow 0} [f(x_0 + h) - 2f(x_0) + f(x_0 - h)] \stackrel{\text{AOL}}{=} 0.$$

Also, since f is twice differentiable, f' is differentiable in a neighborhood of x_0 .

We now compute the limit starting from the left-hand side:

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2} &\stackrel{\frac{0}{0}}{=} \lim_{h \rightarrow 0} \frac{f'(x_0 + h) + f'(x_0 - h) \cdot (-1)}{2h} = \lim_{h \rightarrow 0} \frac{f'(x_0 + h) - f'(x_0 - h)}{2h} \\ &= \lim_{h \rightarrow 0} \frac{f'(x_0 + h) - f'(x_0) + f'(x_0) - f'(x_0 - h)}{2h} \\ &= \lim_{h \rightarrow 0} \left[\frac{f'(x_0 + h) - f'(x_0)}{2h} + \frac{f'(x_0) - f'(x_0 - h)}{h} \right] \\ &\stackrel{(\star)}{=} \frac{f''(x_0)}{2} + \frac{f''(x_0)}{2} = f''(x_0) \end{aligned}$$

where

$$\lim_{h \rightarrow 0} \frac{f'(x_0 - h) - f'(x_0)}{h} \stackrel{t=-h}{=} \lim_{t \rightarrow 0} \frac{f'(x_0 + t) - f'(x_0)}{-t} = \lim_{t \rightarrow 0} - \frac{f'(x_0 + t) - f'(x_0)}{t} \stackrel{\text{AOL}}{=} -f''(x_0). \quad (\star)$$

(b) Prove or disprove the following statement:

If the limit $\lim_{h \rightarrow 0} \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2}$ exists, then f is twice-differentiable at x_0 .

Solution:

False. We choose $f(x) = \begin{cases} \frac{x^2}{2} & x \geq 0 \\ -\frac{x^2}{2} & x < 0 \end{cases}$ for every $x \in \mathbb{R}$ and choose $x_0 = 0$.

We know show that f is not twice differentiable at $x_0 = 0$.

Note that for every $x \in \mathbb{R}$ we have $f(x) + f(-x) = 0$.

It follows that

$$\lim_{h \rightarrow 0} \frac{f(x_0 + h) - 2f(x_0) + f(x_0 - h)}{h^2} = \lim_{h \rightarrow 0} \frac{f(h) + f(-h)}{h^2} = \lim_{h \rightarrow 0} \frac{0}{h^2} = 0.$$

4. (a) Student A has computed $\lim_{x \rightarrow 1} \frac{x^3+3x-4}{x^2-3x+2}$ by using L'Hôpital's rule in the following way:

$$\lim_{x \rightarrow 1} \frac{x^3+3x-4}{x^2-3x+2} = \lim_{x \rightarrow 1} \frac{3x^2+3}{2x-3} = \lim_{x \rightarrow 1} \frac{6x}{2} = 3$$

Explain student A's mistake.

Solution:

Student A used L'Hôpital's rule but the conditions for L'Hôpital's rule don't hold, as the limit $\lim_{x \rightarrow 1} \frac{3x^2+3}{2x-3}$ is not a limit of the form $\frac{0}{0}$ nor of the form $\frac{\infty}{\infty}$ - it is a limit of the form $\frac{6}{-1}$. Student A could have used L'Hôpital's rule only on the first limit, but not on the second one. Indeed, the value of the limit is -6 .

- (b) Student A has proved that every differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ must also be continuously differentiable in the following way:

Let $x_0 \in \mathbb{R}$. Then

$$f'(x_0) \stackrel{\text{def}}{=} \lim_{x \rightarrow x_0} \frac{f(x) - f(x_0)}{x - x_0} \stackrel{\text{L'Hôpital's rule}}{=} \lim_{x \rightarrow x_0} \frac{f'(x)}{1} = \lim_{x \rightarrow x_0} f'(x)$$

Explain student A's mistake.

Solution:

Student A assumed that $\lim_{x \rightarrow x_0} f'(x)$ exists in the extended sense, which is one of the assumptions of L'Hôpital's rule,

but this assumption doesn't always hold. Indeed, the function defined by $f(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$ for every $x \in \mathbb{R}$

was seen in class as a function that is not continuously differentiable at $x_0 = 0$, because $\lim_{x \rightarrow x_0} f'(x)$ does not exist in the extended sense.

- (c) Student A found a counter-example for Darboux's theorem: Consider the function defined by

$$f(x) = \begin{cases} x & x > 0 \\ -x & x < 0 \end{cases}$$

Then f is differentiable by AOD. Note that $f'(1) = 1$ and $f'(-1) = -1$, but there exists no $c \in [-1, 1]$ for which $f'(c) = 0$.

Explain student A's mistake.

Solution:

The conditions for Darboux's theorem do not hold; the function is not differentiable on $[-1, 1]$ as it is not defined at $x_0 = 0$, and therefore not differentiable at $x_0 = 0$.

5. Let f be a function that is defined on $(0, 2)$. Suppose that f is differentiable.

- (a) Suppose that $f'(x) \cdot f'\left(\frac{1}{x}\right) < 0$ for every $1 \neq x \in (0, 2)$.

Prove that $f'(1) = 0$.

Solution:

First note that as $f'(x) \cdot f'\left(\frac{1}{x}\right) < 0$ for every $1 \neq x \in (0, 2)$, then $f'(x) \neq 0$ for every $1 \neq x \in (0, 2)$.

As $f'\left(\frac{2}{3}\right) \cdot f'\left(\frac{3}{2}\right) < 0$, then by Darboux's theorem there exists a $\frac{2}{3} < c < \frac{3}{2}$ for which $f'(c) = 0$.

Thus, $c = 1$.

- (b) Suppose that $\lim_{x \rightarrow 2^-} f(x)$ does not exist in the extended sense. Prove that there exists a $c \in (0, 2)$ for which $f'(c) = 0$.

Solution:

ATC that $f'(x) \neq 0$ for every $0 < x < 2$.

Suppose first that $f'(x) > 0$ for every $x \in (0, 2)$. Then f is strictly increasing, and therefore $\lim_{x \rightarrow 2^-} f(x)$ exists in the extended sense and

$$\lim_{x \rightarrow 2^-} f(x) = \sup \{f(x) : 0 < x < 2\}$$

Contradiction.

Suppose now that $f'(x) < 0$ for every $x \in (0, 2)$. Then f is strictly decreasing, and therefore $\lim_{x \rightarrow 2^-} f(x)$ exists in the extended sense and

$$\lim_{x \rightarrow 2^-} f(x) = \inf \{f(x) : 0 < x < 2\}$$

Contradiction.

We conclude that there exists two points $0 < x_1 < x_2 < 2$ for which $f'(x_1) \cdot f'(x_2) < 0$. As f is differentiable on $[x_1, x_2]$, then by Darboux's theorem there exists an $x_1 < x < x_2$ for which $f'(x) = 0$. Contradiction.

(c) Student X was kind of impressed when she saw the proof that $\sin(x) < x$ for every $0 < x < \frac{\pi}{2}$ in the first semester.

Feeling powerful, she wanted to prove an inequality on her own: an inequality that would be both more impressive and more accurate. She asked her TA for help.

Prove that there exists a $\delta > 0$ such that for every $0 < x < \delta$ we have

$$x - 0.17x^3 < \sin(x) < x - 0.16x^3$$

Solution:

We compute

$$\lim_{x \rightarrow 0^+} \frac{\sin(x) - x}{x^3} \stackrel{\frac{0}{0}}{=} \lim_{x \rightarrow 0^+} \frac{\cos(x) - 1}{3x^2} \stackrel{-\frac{1}{2} \text{ famous}}{=} -\frac{1}{6}$$

As $\lim_{x \rightarrow 0^+} \frac{\sin(x) - x}{x^3} = -\frac{1}{6}$, then there exists a $\delta_1 > 0$ such that for every $0 < x < \delta_1$ we have

$$-0.17 < \frac{\sin(x) - x}{x^3} < -0.16 \quad (\star)$$

Choose $\delta = \delta_1$ and let $0 < x < \delta$. Then $(\star)$ hold for that x .

Multiplying by x^3 we obtain $x - 0.17x^3 < \sin(x) < x - 0.16x^3$.

6. Let $L \in \mathbb{R}$.

(a) Let $g_1 : \mathbb{R} \rightarrow \mathbb{R}$. Suppose g_1 is an odd function, i.e., $g_1(-x) = -g_1(x)$ for every $x \in \mathbb{R}$.

Suppose in addition that $\lim_{x \rightarrow \infty} g_1(x) = L$. Prove that $\lim_{x \rightarrow -\infty} g_1(x) = -L$.

Solution:

We compute

$$\lim_{x \rightarrow -\infty} g_1(x) \stackrel{t=-x}{=} \lim_{t \rightarrow \infty} g_1(-t) \stackrel{\text{odd}}{=} \lim_{t \rightarrow \infty} -g_1(t) \stackrel{\text{AOL}}{=} -L$$

(b) Let $g_2 : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function.

Prove that if $\lim_{x \rightarrow \infty} g_2(x) = \lim_{x \rightarrow -\infty} g_2(x) = L$, then g_2 has an extreme point in $\mathbb{R}$.

Solution:

First, if g_2 is constant then every point is extreme. We now assume g_2 is not constant and consider the function defined by

$$\Psi(x) = \begin{cases} g_2(\tan(x)) & -\frac{\pi}{2} < x < \frac{\pi}{2} \\ L & x = \frac{\pi}{2} \\ L & x = -\frac{\pi}{2} \end{cases}, \quad \forall x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

Note that as g_2 is continuous and as $\lim_{x \rightarrow \infty} g_2(x) = \lim_{x \rightarrow -\infty} g_2(x) = L$, then

$$\lim_{x \rightarrow \frac{\pi}{2}^-} \Psi(x) = \lim_{x \rightarrow \frac{\pi}{2}^-} g_2(\tan(x)) \stackrel{y=\tan(x)}{=} \lim_{y \rightarrow \infty} g_2(y) = L = \Psi\left(\frac{\pi}{2}\right)$$

and similarly,

$$\lim_{x \rightarrow -\frac{\pi}{2}^+} \Psi(x) = \lim_{x \rightarrow -\frac{\pi}{2}^+} g_2(\tan(x)) \stackrel{y=\tan(x)}{=} \lim_{y \rightarrow -\infty} g_2(y) = L = \Psi\left(-\frac{\pi}{2}\right)$$

It follows that Ψ is left-continuous at $\frac{\pi}{2}$ and right-continuous at $-\frac{\pi}{2}$. As g_2 is continuous, then Ψ is also continuous on $(-\frac{\pi}{2}, \frac{\pi}{2})$. Thus Ψ is continuous on $[-\frac{\pi}{2}, \frac{\pi}{2}]$. It follows that by Weirstrass's theorem, Ψ attains a maximum and a minimum on $[-\frac{\pi}{2}, \frac{\pi}{2}]$, namely, there exist $x_1, x_2 \in [-\frac{\pi}{2}, \frac{\pi}{2}]$ such that

$$\Psi(x_2) \leq \Psi(x) \leq \Psi(x_1), \quad \forall x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

As g_2 is not constant, then there exist $y_1, y_2 \in \mathbb{R}$ such that $y_1 \neq y_2$ and $g(y_1) \neq g(y_2)$.

We note that there exist $x_1 = \arctan(y_1)$, $x_2 = \arctan(y_2)$. We note that $x_1, x_2 \in (-\frac{\pi}{2}, \frac{\pi}{2})$.

As $y_1 \neq y_2$, then $x_1 \neq x_2$. Thus

$$\Psi(x_1) = g_2(\tan(x_1)) = g_2(\tan(\arctan(y_1))) = g_2(y_1) \neq g_2(y_2) = g_2(\tan(\arctan(y_2))) = g_2(\tan(x_2)) = \Psi(x_2)$$

Thus Ψ is not constant, and as $\Psi(-\frac{\pi}{2}) = \Psi(\frac{\pi}{2})$, at least one of these extreme points lies in the interior of the interval, namely $\{x_1, x_2\} \neq \{-\frac{\pi}{2}, \frac{\pi}{2}\}$. Denoting this point by x_0 , and defining $y_0 = \tan(x_0)$ (well-defined as $x_0 \in (-\frac{\pi}{2}, \frac{\pi}{2})$) then y_0 is an extreme point of g_2 .

(c) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function.

Consider the function defined by $g(x) = f(x) - f(-x)$ for every $x \in \mathbb{R}$. Suppose that $\lim_{x \rightarrow \infty} g(x) = 0$.

Conclude, using items (a) and (b), that there exists a $c \in \mathbb{R}$ for which $g'(c) = 0$.

Solution:

We first show that g is an odd function: let $x \in \mathbb{R}$. We note that

$$g(-x) = f(-x) - f(-(-x)) = f(-x) - f(x) = -(f(x) - f(-x)) = -g(x)$$

Since we assumed $\lim_{x \rightarrow \infty} g(x) = 0$ then by item (a) we also obtain $\lim_{x \rightarrow -\infty} g(x) = -0 = 0$.

As f is differentiable, then by AOD g is differentiable and therefore continuous and therefore by item (b) there exists an $c \in \mathbb{R}$ such that c is an extreme point. As g is differentiable, then by Fermat's theorem we have $g'(c) = 0$.

(d) Prove that there exists a $d \in \mathbb{R}$ for which $f'(d) = 0$.

Solution:

By AOD, g is differentiable

$$g'(x) = f'(x) - f'(-x) \cdot (-1) = f'(x) + f'(-x), \quad \forall x \in \mathbb{R}$$

As $g'(c) = 0$, then $f'(c) + f'(-c) = 0 \Rightarrow f'(c) = -f'(-c)$.

If $c = -c$ then $c = 0$ and therefore $f'(0) = -f'(0) \Rightarrow f'(0) = 0$.

We choose $d = 0$ and complete the proof.

Otherwise, $c, -c$ are two different points. As $f'(-c) = -f'(c)$, then

$$f'(c) \cdot f'(-c) = -[f'(c)]^2 \leq 0$$

Thus, by the generalized Darboux theorem, there exists a $d \in \mathbb{R}$ between $-c$ and c for which $f'(d) = 0$.